Design and Implementation of an ECG Analog Front-End

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Abstract— This paper presents the design, simulation, and PCB layout of an Analog Front-End (AFE) for electrocardiographic (ECG) signal acquisition, in compliance with IEC 60601-1 and IEC 60601-2-25 safety and performance standards. The proposed architecture features a robust frontend protection stage capable of withstanding ESD and defibrillation pulses. The signal chain comprises an instrumentation amplifier (INA) and a Right Leg Drive (RLD) circuit to enhance Common-Mode Rejection Ratio (CMRR). The analog processing stage is optimized for a 0.05 Hz to 150 Hz bandwidth, incorporating high-pass, low-pass, and a 50 Hz notch filter for power-line interference attenuation. Performance metrics were obtained via LTspice simulation and Monte Carlo analysis to assess component tolerance effects. The results confirm the system's viability for clinical grade biopotential monitoring while ensuring patient safety. The system targets an accessible price point, positioning it as a viable solution for home-based cardiac monitoring.

Keywords— ECG, AFE, IEC 60601, LTspice, RLD, PCB.

I. INTRODUCTION

Cardiovascular diseases remain the leading cause of global mortality [1], necessitating reliable and accessible diagnostic tools. The ECG is the primary instrument for cardiac monitoring, yet its effectiveness depends on the precise acquisition of biopotentials in the millivolt range [2]. These signals are inherently susceptible to significant interference from external environmental noise, power-line coupling, and physiological artifacts such as respiration or patient movement [3]. To ensure clinical-grade reliability, the AFE must comply with IEC 60601-1 and IEC 60601-2-25. This paper presents a design centered on an INA and RLD architecture, utilizing an analog processing chain tuned to a 0.05 Hz – 150 Hz bandwidth, as illustrated in Fig. 1. The work aims to validate the transition from LTspice models to a functional PCB through a comparative performance analysis.

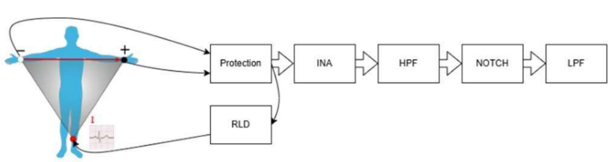

Fig. 1. Block diagram of the ECG AFE signal chain

II. CIRCUIT DESIGN

A. Power Supply and Voltage Reference

The AFE is powered via USB-C, which supplies 5V. Since USB power rails can introduce significant noise, an ADP151 low-dropout (LDO) regulator is used to regulate the supply down to a clean 3.3V rail, benefiting from its low noise output and high PSRR. A virtual ground is generated using a voltage divider buffered by an AD8534 operational amplifier. This configuration provides a stable 1.65V, which is essential for maintaining the signal within the linear operating range of the single-supply analog chain.

B. Input Protection and Instrumentation Stage

The electrode interface incorporates a two-stage protection circuit. The first stage uses SMBJ90A TVS diodes to clamp high-voltage transients, such as defibrillation pulses, preventing them from reaching the sensitive analog stages. The second stage consists of current-limiting resistors and BZX84C3V3 Zener diodes, which clamp any remaining transients to 3.3V. The differential signal is then acquired by a MAX41400 instrumentation amplifier, selected for its high CMRR (120 dB typical, 106 dB minimum) and low-noise performance, while remaining cost-effective. This stage provides an initial gain of 10 V/V. To further suppress common-mode interference, an active RLD circuit senses the common-mode level and drives a compensatory signal back to the patient. The current injected into the patient through the RLD path is limited to below 10 µA, in compliance with the IEC 60601-1 [4] patient leakage current requirements.

C. Multistage Signal Conditioning and Gain Distribution

The signal is processed through several stages to build up the gain without distorting the signal or hitting saturation. This modular approach makes it easier to tune each filter and keep the noise low:

1) High-Pass Filtering (HPF): A 0.05 Hz stage eliminates DC offsets and baseline wander [3]. This stage incorporates an additional gain of 10 V/V, bringing the cumulative gain to 100 V/V while stabilizing the signal baseline.

2) Active Notch Filter: This stage is used to cut out the 50 Hz power-line hum that usually interferes with the ECG signal. By removing this noise, ensuring the QRS complex remains clearly distinguishable and easy to see without unwanted oscillations.

3) Low-Pass Filtering (LPF): A 150 Hz low-pass stage removes high-frequency muscular artifacts and electromagnetic interference [3]. This final stage provides a gain of 5 V/V, resulting in a total system gain of 500 V/V (approx. 54 dB).

III. PCB IMPLEMENTATION

The circuit was implemented on a two-layer FR4 PCB using 0603 SMD components, keeping the board compact and the overall system cost low, consistent with the goal of an affordable home-use device. A four-layer stackup was considered but excluded, as two layers proved sufficient for the signal chain requirements. The layout follows the signal chain order, from input protection to the final output stage, to minimize inter-stage interference.

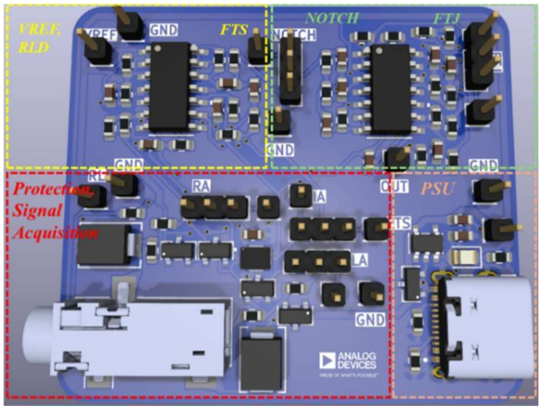

Fig. 2. PCB layout with annotated functional blocks.

Decoupling capacitors are placed close to each IC's supply pins. Three-pin test headers are placed at each block boundary, allowing any stage to be driven independently by a signal generator and characterized separately, useful for validating individual filter responses before full-chain testing.

IV. SIMULATION AND TOLERANCE ANALYSIS

To bridge the gap between ideal simulation and physical implementation, a Monte Carlo Analysis was performed. This study evaluates how 0.1% resistor and 10% capacitor tolerances affect the AFE's performance. Since a physical prototype is subject to these real-world variations, this analysis ensures the design remains robust even under unfavorable component combinations.

The values presented in Table I were obtained by running 50 iterations in LTspice. This provides a realistic expectation of how the circuit will behave in real life.

Table I. Performance Variations under Component Tolerances

Param.	Values		
	Simulated	Monte Carlo	Deviation [%]
Total Gain (V/V)	500	504.5-510	+0.9, +2
HPF cutoff @-3dB (Hz)	0.05	0.0462-0.0535	-7.6, +7.0
LPF cutoff @-3dB (Hz)	150	112.22 -177.75	-25.2, +18.4
Notch atten. @50 Hz (dB)	-40	-19.34	51.6
Notch center freq. (Hz)	50	47.5– 56.2	-5, +12.4

V. CONCLUSION AND FUTURE WORK

This work demonstrates that a clinical-grade ECG signal acquisition front-end can be realized at low cost using off-the shelf components, making it a viable foundation for accessible cardiac monitoring. The design goals were successfully met. A functional ECG Analog Front-End was designed and simulated, targeting implementation on a compact two-layer PCB, achieving the target gain and bandwidth while maintaining patient safety compliance. The total BOM cost is approximately 50 RON (~10 EUR) per unit at 1000-unit volume pricing, excluding assembly, confirming the system's viability as a low-cost solution. The modular architecture, with individually testable stages, simplifies both validation and future iterations of the design. Monte Carlo analysis revealed that the system is robust overall, apart from the Twin-T notch filter, which shows significant sensitivity to component tolerances and remains the main limitation of the current design. Addressing this through tighter component selection or an alternative topology will be a priority in future revisions.

Future work includes physical PCB assembly and experimental validation of the simulated results. A subsequent development phase will integrate an ADC and wireless transmission module, enabling real-time cardiac data transfer to a mobile device and completing the transition toward a fully home-based diagnostic tool.

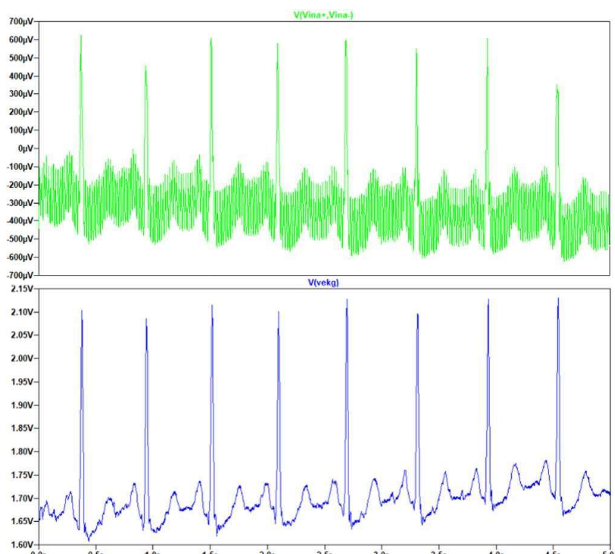

Fig. 3. ECG signal at INA input (top) and AFE output (bottom).

[1] M. Lindstrom et al., "Global burden of cardiovascular diseases and risks collaboration, 1990–2021," J. Am. Coll. Cardiol., vol. 80, no. 25, pp. 2372–2425, Dec. 2022.

[2] M. Hammad et al., "Detection of abnormal heart conditions based on characteristics of ECG signals," Measurement, vol. 125, pp. 634–644, 2018.

[3] S. Luo and P. Johnston, "A review of electrocardiogram filtering," J. Electrocardiol., vol. 43, no. 6, pp. 486–496, 2010.

[4] International Electrotechnical Commission, "Medical electrical equipment – Part 1: General requirements for basic safety and essential performance," IEC 60601-1, Ed. 3.1, 2012.